

## Benchmark với các threshold của 2 mô hình :

### 1. ATE Model

**ATE Model — Span Threshold Sweep (Gold Test)**

| Span Threshold | Accuracy | Precision | Recall | F1     |
|----------------|----------|-----------|--------|--------|
| 0.0000         | 0.6112   | 0.6578    | 0.6037 | 0.6296 |
| 0.1000         | 0.6112   | 0.6578    | 0.6037 | 0.6296 |
| 0.2000         | 0.6112   | 0.6578    | 0.6037 | 0.6296 |
| 0.3000         | 0.6112   | 0.6578    | 0.6037 | 0.6296 |
| 0.4000         | 0.6112   | 0.6578    | 0.6037 | 0.6296 |
| 0.5000         | 0.6112   | 0.6600    | 0.6037 | 0.6306 |
| 0.6000         | 0.6225   | 0.6851    | 0.5869 | 0.6322 |
| 0.7000         | 0.6212   | 0.7115    | 0.5640 | 0.6293 |
| 0.8000         | 0.6150   | 0.7326    | 0.5305 | 0.6154 |
| 0.9000         | 0.6162   | 0.7672    | 0.4924 | 0.5998 |

### 2. Gate Classifier

**Gate Classifier — Threshold Sweep (Gold Test)**

| Threshold | Accuracy | Precision | Recall | F1     |
|-----------|----------|-----------|--------|--------|
| 0.0000    | 0.5675   | 0.5675    | 1.0000 | 0.7241 |
| 0.1000    | 0.8350   | 0.8084    | 0.9295 | 0.8648 |
| 0.2000    | 0.8387   | 0.8143    | 0.9273 | 0.8671 |
| 0.3000    | 0.8400   | 0.8171    | 0.9251 | 0.8678 |
| 0.4000    | 0.8425   | 0.8203    | 0.9251 | 0.8696 |
| 0.5000    | 0.8438   | 0.8219    | 0.9251 | 0.8705 |
| 0.6000    | 0.8450   | 0.8248    | 0.9229 | 0.8711 |
| 0.7000    | 0.8462   | 0.8290    | 0.9185 | 0.8715 |
| 0.8000    | 0.8438   | 0.8297    | 0.9119 | 0.8688 |
| 0.9000    | 0.8475   | 0.8360    | 0.9097 | 0.8713 |